DUNE Near Detector Supernova Sensitivity Study

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Abstract

This study investigates the sensitivity of the Deep Underground Neutrino Experiment (DUNE) near detector complex to a Galactic supernova neutrino burst, with particular emphasis on The Muon Spectrometer (TMS). This study makes use only of publicly available information. SNOwGLoBES is used to estimate event rates, while MARLEY and Geant4 are combined to model the detector response. For a supernova at a distance of 10 kpc, approximately 10.5 TMS events are expected for the Garching flux model and 52.3 events for the Livermore flux model. After applying simple cuts to suppress the cosmic-muon background, 8.3 supernova neutrino events and 17.8 cosmic-muon background events remain. The study relies on simplifying assumptions and therefore provides a first-order estimate. A more realistic assessment would require detailed background, trigger and reconstruction models.

1 Introduction

A core-collapse supernova releases almost all of the gravitational binding energy in the form of neutrinos, producing an intense burst with characteristic energies of $\mathcal{O}(10\,\mathrm{MeV})$ over a timescale of several seconds [1]. SN 1987A remains the only supernova from which neutrinos have been detected directly, with the signal observed by several experiments [1]. Since then, many new neutrino detectors have been built. If a supernova occurs in our Galactic neighborhood, modern neutrino experiments are expected to detect a large number of neutrino events. Such a detection would provide valuable information on both supernova physics and neutrino properties. The Deep Underground Neutrino Experiment (DUNE) will be a state-of-the-art neutrino observatory. It comprises a 40 kt liquid argon detector as the far detector and multiple detector components at the DUNE near detector complex [2]. The supernova-neutrino sensitivity of the DUNE far detector has already been studied [3]. Although the DUNE near detector complex is designed primarily for accelerator-neutrino measurements, its variety of target materials may provide additional sensitivity to a supernova neutrino burst. This study estimates the supernova-neutrino event rates in the DUNE near detector complex. Particular emphasis is placed on the TMS, for which interaction-specific detector response matrices are obtained using MARLEY and Geant4 and incorporated into SNOwGLoBES. The simulation analysis code developed for this project is available on GitHub [4].

2 Supernova Neutrino Signal Simulation with SNOwGLoBES

The open-source software SNOwGLoBES is used to simulate the supernova-neutrino signal in a given detector for a supernova at a distance of 10 kpc [5]. The calculations are based on the supernova flux model, material-specific cross sections, detector-specific smearing matrices, and post-smearing efficiencies. In this study, both the Livermore flux model and the Garching flux model are used [6, 7]. SNOwGLoBES uses 200 equally spaced reconstructed energy bins in the range $0.5\,\mathrm{MeV} \leq E_{\mathrm{reco}} \leq 100\,\mathrm{MeV}$. This corresponds to an energy bin width of 0.4975 MeV [5]. To ensure uniform energy binning, this system is used throughout the whole study. The SNOwGLoBES smearing matrix represents the reconstructed-energy probability distribution for a fixed true neutrino energy.

$$R_{ij,c} = P(E^i_{reco} | E^j_{true,\nu}) \tag{1}$$

The superscript index i represents the i-th energy bin in the SNOwGLoBES energy grid. The row index i corresponds to the reconstructed energy bin, whereas the column index j corresponds to the true neutrino-energy bin. Each matrix element R_{ij} describes the probability of a reconstructed-energy measurement in energy bin i for a true neutrino energy in energy bin j for an interaction channel c.

3 DUNE Near Detector site

The DUNE near detector complex consists of three main detector components: ND-LAr, TMS, and SAND. Additionally, SAND comprises three components: GRAIN, ECAL, and STT [2].

Detector	Material	Mass [t]	References
ND-LAr	Argon	67	[2]
TMS	Scintillator	35	[8]
	Steel	750	[8]
GRAIN	Argon	1	[9]
ECAL	Lead	88.7	[9, 10]
	Scintillator	9.2	[9, 10]
STT	CH_2	4.7	[2]

Table 1: Materials and masses for the detector components at the DUNE near detector site. Masses are given in tons.

3.1 Implementation of the DUNE Near Detector in SNOwGLoBES

Each detector component is defined in the SNOwGLoBES detector configuration file by its target mass in kt and target normalization. The target normalization gives the number of reference targets per atomic mass unit for the respective detector material. For example, for ^{40}Ar detectors, the reference material is argon, so the target normalization is $1/40 = 0.025$. Detector components containing multiple target materials are represented by separate SNOwGLoBES detector configurations.

3.2 Implementation of ^{56}Fe interaction channels

As a simplifying approximation, the 750 t steel absorber mass of the TMS is modeled as pure ^{56}Fe. Contributions from other constituents of the steel are neglected. However, SNOwGLoBES does not support iron as a detector material natively. Implementing ^{56}Fe requires a dedicated channel file. The implemented channels include charged-current interactions of ν_e and $\bar{\nu}_e$, neutral-current interactions of all neutrino flavors, and elastic neutrino-electron scattering.

Interaction Channel	Reaction	Weight
CC ν_e	$\nu_e + {}^{56}\mathrm{Fe} \to \mathrm{e}^- + {}^{56}\mathrm{Co}^*$	1
CC $\bar{\nu}_e$	$\bar{\nu}_e + {}^{56}\mathrm{Fe} \to \mathrm{e}^+ + {}^{56}\mathrm{Mn}^*$	1
NC ν_x	$\nu_x + {}^{56}\mathrm{Fe} \to \nu_x + {}^{56}\mathrm{Fe}^*$	1
eES ν_x	$\nu_x + \mathrm{e}^- \to \nu_x + \mathrm{e}^-$	26

Table 2: Neutrino interaction channels on ^{56}Fe implemented in SNOwGLoBES, including the corresponding channel weight. Here, ν_x represents all neutrino and antineutrino flavors.

A SNOwGLoBES channel file also contains a corresponding channel weight for each channel. For example, since ^{56}Fe is the reference target and ^{56}Fe contains 26 electrons, the eES channel is assigned a weight of 26.
Material-specific cross sections are supplied separately for each interaction channel. Ref. [11] provides

the charged-current cross sections over the relevant supernova-neutrino energy range. For neutrino energies in the range of 7.5 MeV $\leq E_\nu \leq$ 100 MeV, cross sections forν_e and $\bar{\nu}_e$ channels are given in the paper. For neutral current interactions, cross sections from Ref. [12] are used. It provides NC cross sections on ^{56}Fe for 10 MeV $\leq E_\nu \leq$ 100 MeV. The tabulated cross sections are interpolated onto the SNOwGLoBES energy grid using a shape-preserving PCHIP interpolator (Piecewise Cubic Hermite Interpolating Polynomial). For neutrino energies lower than the first cross-section value in the references, linear extrapolation is used. Calculated negative cross-section values are set to zero. As a simplifying assumption, the same NC cross section is applied to all neutrino and antineutrino flavors. Cross sections for elastic neutrino-electron scattering are included with the SNOwGLoBES package.

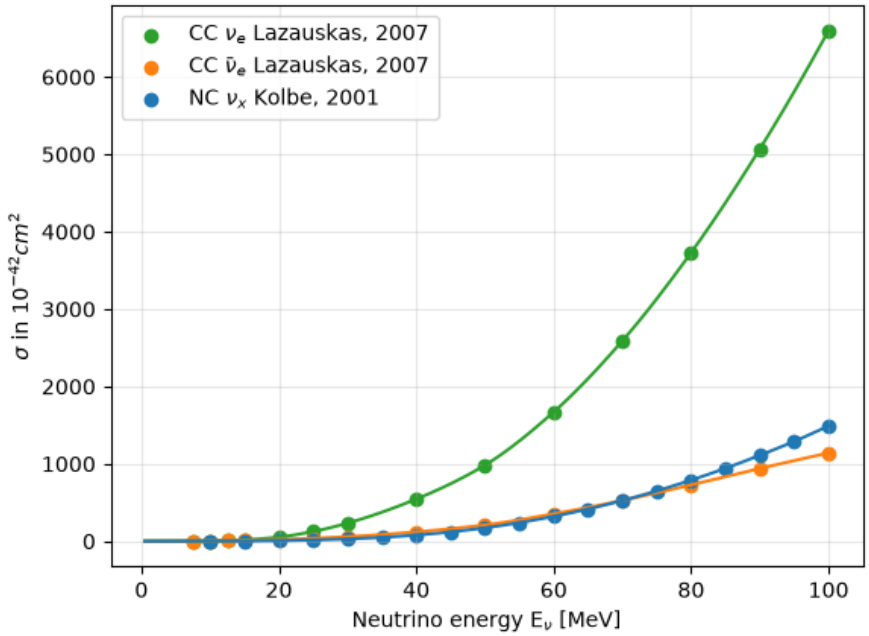


Figure 1: Implemented ^{56}Fe cross sections for SNOwGLoBES. Markers show data points from the corresponding references [11, 12]. The continuous lines show PCHIP interpolation and linear extrapolation.

4 Supernova Signal calculation

In this study, two supernova neutrino flux models are used: the Livermore [6] and Garching [7] models. First, the unsmeared event rate for every detector is calculated. The calculations are based on the detector masses specified in Table 1. The material of the STT is approximated as scintillator in SNOwGLoBES. For the expected unsmeared event rate of a given channel c, SNOwGLoBES calculates:

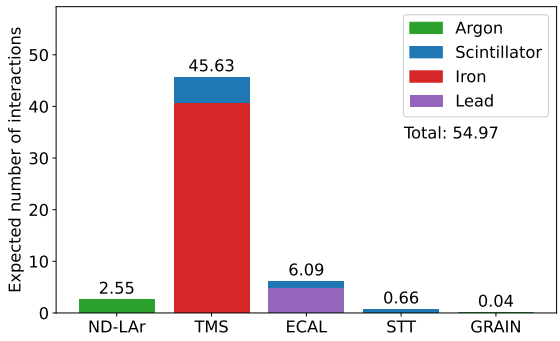
$$\frac{dN_c}{dE_\nu} = N_T \omega_c \Phi(E_\nu) \sigma_c(E_\nu) \tag{2}$$

N_T represents the number of reference targets and ω_c the corresponding channel weight. $\Phi(E_\nu)$ is the SN neutrino fluence, and $\sigma_c(E_\nu)$ describes the channel-specific cross section of the detector material. This neglects the detector response and computes the number of expected SN neutrino events without accounting for detector geometry.

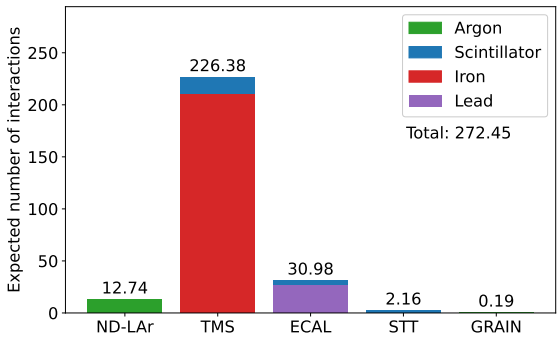
The expected unsmeared event rate for the j-th energy bin can be written as:

$$N^j_{\text{unsmeared, c}} = \int_{E_j^-}^{E_j^+} N_T \omega_c \Phi(E_\nu) \sigma_c(E_\nu) dE_\nu \tag{3}$$

E_j^- represents the lower bin edge of the j-th energy bin, while E_j^+ represents the upper bin edge.



(a) Garching flux model. 10 kpc SN distance.



(b) Livermore flux model. 10 kpc SN distance.

Figure 2: Expected unsmeared supernova-neutrino interaction rates in the DUNE near detector complex for the Garching and Livermore flux models.

5 TMS Detector Model

5.1 Geant4 Model

A simplified model of the TMS was implemented in Geant4 [13]: 34 layers of 15 mm-thick iron plates, followed by 22 40 mm-thick and 24 80 mm-thick iron plates [14]. Between adjacent iron plates, a 16 mm scintillator plate is positioned with a 17 mm air gap on each side. Additionally, each layer is 7 m wide and 3.2 m tall.

For each event in the TMS, Geant4 stores the deposited energy E_{scint} and the total path length L_{scint} inside the scintillator. The goal of the Geant4 model is to calculate realistic detector response matrices for each neutrino interaction channel. These matrices are then transferred to SNOwGLoBES to calculate the supernova signal.

5.2 Detector response matrices with Geant4

For the charged-current interaction on ^{56}Fe, the outgoing electron energy E_e can be described as follows under the first simplifying assumption:

$$E_e = E_\nu - Q \tag{4}$$

The Q-value represents the energy difference between the initial and final states of a nuclear reaction. For ν_e CC, $Q = 4.56\,\text{MeV}$, and for $\bar{\nu}_e$ CC, $Q = 4.71\,\text{MeV}$ [15]. For the initial position of the electron in the Geant4 TMS model, a random iron layer is chosen, weighted by the total mass of the iron plate. Then, a random point in the iron is selected as the Geant4 electron initialization position. For every generated electron, a random direction is generated so that the electrons are distributed uniformly and isotropically in the detector.

The midpoint of a SNOwGLoBES energy bin is chosen as the initial neutrino energy, and the electron energy is calculated with (4). For every neutrino energy, 100,000 events are generated. The deposited energy in the TMS scintillator is stored for every event and neutrino energy. Using (1), the SNOwGLoBES smearing matrix files are created. Each column of the smearing matrix is not necessarily normalized to unity. Events depositing less than 0.5 MeV in the scintillator fall outside the reconstructed energy range used by SNOwGLoBES and therefore do not contribute to the matrix. Consequently, the column sum represents the fraction of generated events entering the considered reconstructed-energy range.

5.3 MARLEY integration

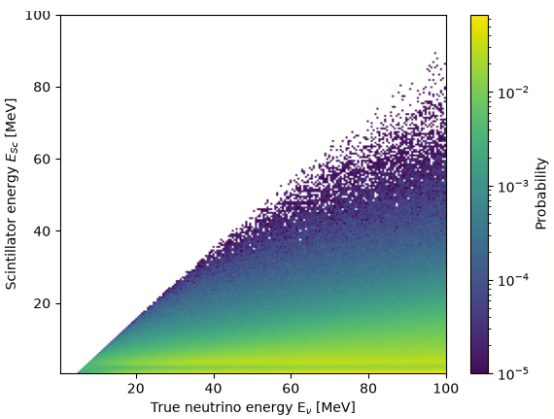
MARLEY is an open-source Monte Carlo neutrino event generator [16]. It mainly focuses on the low-energy regime, which is important for supernova neutrinos. Although the central purpose of MARLEY

is to simulate neutrino events on argon, reaction files for other materials can also be created and used with MARLEY's neutrino interaction simulation. In this study, MARLEY is used to calculate the primary particles with their corresponding energies and directions. This information is then transferred to the Geant4 TMS model, and a random position in a randomly chosen iron plane is selected. The two software tools were connected using the Geant4 files provided in the MARLEY package. The combination of MARLEY and Geant4 replaces the simplifying assumptions of Eq. (4).

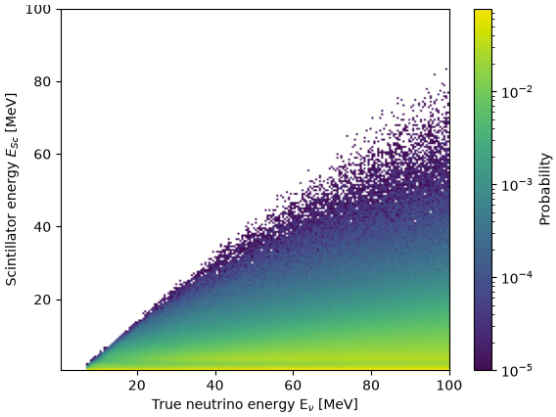
5.4 Detector response matrices for ^{56}Fe

5.4.1 Charged-current interaction

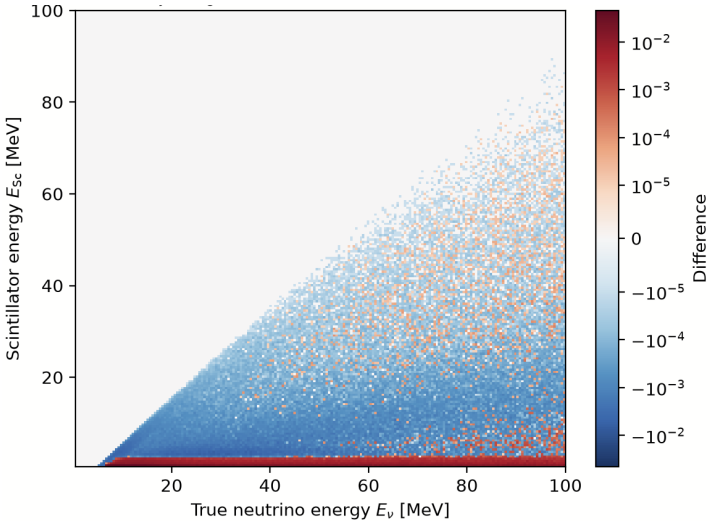
The MARLEY reaction file for the ν_e CC reaction on iron was originally created by Sam Hedges in 2020. It is based on data from J. Rapaport et al. [17] and N. Paar et al. [18]. For excitation energies below 15.7 MeV, the reaction file contains experimental data, while for energies above 15.7 MeV theoretical RQRPA results were used.



(a) Simplified detector-response model.



(b) MARLEY combined with Geant4.



(c) Difference $R_{\mathrm{MARLEY}} - R_{\mathrm{simplified}}$.

Figure 3: Comparison of the simplified and MARLEY-based detector response for the ν_e CC interaction channel.

As seen in Figure 3c, combining MARLEY and Geant4 and generating more realistic primary particles leads to a higher probability of energy deposition in lower-energy regimes. One possible explanation is that the same available energy is distributed among more particles, which leads to lower deposits on average.

For the $\bar{\nu}_e$ CC interaction on ^{56}Fe, a MARLEY reaction file was created using the experimental Gamow–Teller strength distribution measured in Ref. [19]. The data were digitized from Fig. 3a of the publication. Below the particle-emission threshold, the strengths were assigned to known 1^+ states of ^{56}Mn, with strengths distributed equally if multiple states occurred within one experimental energy bin. Bins without matching states were omitted. Above the threshold, the original excitation-energy bins were retained.

5.4.2 Neutral-current interactions

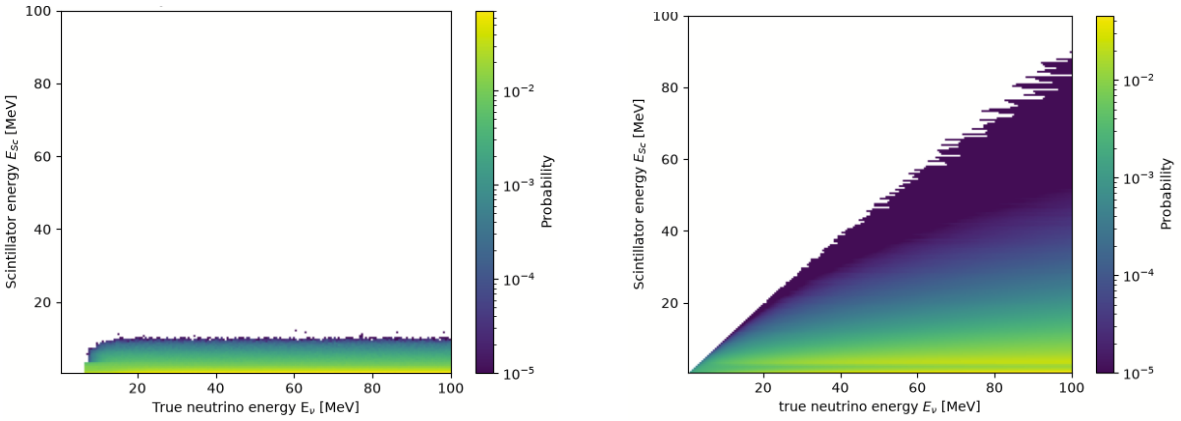
For neutral-current interactions on iron, a MARLEY reaction file originally created by Sam Hedges was used. It is based on Ref. [20]. It uses experimental data for excitation energies below 10 MeV and shell model calculations for higher energies. Similar to the CC ν_e smearing matrix generation, the MARLEY primary particle information for the NC interaction is transferred to the Geant4 model. 100,000 neutrino events per 0.4975 MeV energy bin were generated, and the smearing matrix was calculated. As a simplifying assumption, it is assumed that the NC smearing matrices are the same for all neutrino and antineutrino flavors.

5.4.3 Neutrino-electron elastic scattering

MARLEY was not used to calculate neutrino-electron elastic scattering smearing matrices. Instead, the neutrino-electron elastic scattering smearing matrix was created by folding the detector response matrix with the electron energy distribution matrix:

$$R_{\text{eES}} = R_{\text{response}} \cdot R_{\text{e-distr.}} \tag{5}$$

The detector response matrix was created by generating electrons randomly in an iron layer. The direction of the outgoing electron is sampled isotropically and uniformly since this study does not assume a fixed supernova direction. Instead, it calculates estimates for a supernova averaged over all possible SN directions. The electron energy distribution after an elastic scattering event is provided by SNOwGLoBES. The final smearing matrix is calculated using Eq. (5).



(a) Neutral-current interactions on iron.

(b) Neutrino-electron elastic scattering on iron.

Figure 4: Smearing matrices for two neutrino interaction channels in the TMS.

5.5 Detector response matrices for Scintillator

In SNOwGLoBES, several interaction channels on scintillator are implemented: inverse beta decay (IBD), neutrino-elastic scattering, $\nu_e/\bar{\nu}_e$ CC, and ν_x NC on both ^{12}C and ^{13}C.

The main contributing interaction channels on scintillator are IBD, NC on ^{12}C, and neutrino-electron elastic scattering. These are the only scintillator channels considered in this study. Calculating the eES smearing matrix for scintillator is identical to the process of generating the eES iron smearing matrix. The only difference is that the electrons are generated in a scintillator layer instead of an iron layer.

5.5.1 Inverse beta decay interaction

IBD describes the interaction of a $\bar{\nu}_e$ with a proton:

$$\bar{\nu}_e + p \rightarrow e^+ + n \tag{6}$$

Neglecting neutron recoil, the positron energy is approximated as $E = E_\nu - 1.804\,\text{MeV}$ [21]. For 100,000 events per neutrino energy bin, the positron is generated in a random scintillator layer in Geant4. The distribution of deposited energy in the scintillator is used to calculate the SNOwGLoBES smearing matrix (Eq. (1)).

5.5.2 Neutral current channels

For NC interactions on ^{12}C, the deexcitation energy distribution after the neutrino interaction has a strong peak at 15.11 MeV [22]. As an approximation, a 15.11 MeV γ was generated isotropically in a random scintillator layer. The energy-deposition distribution was used to create the SNOwGLoBES smearing matrix.

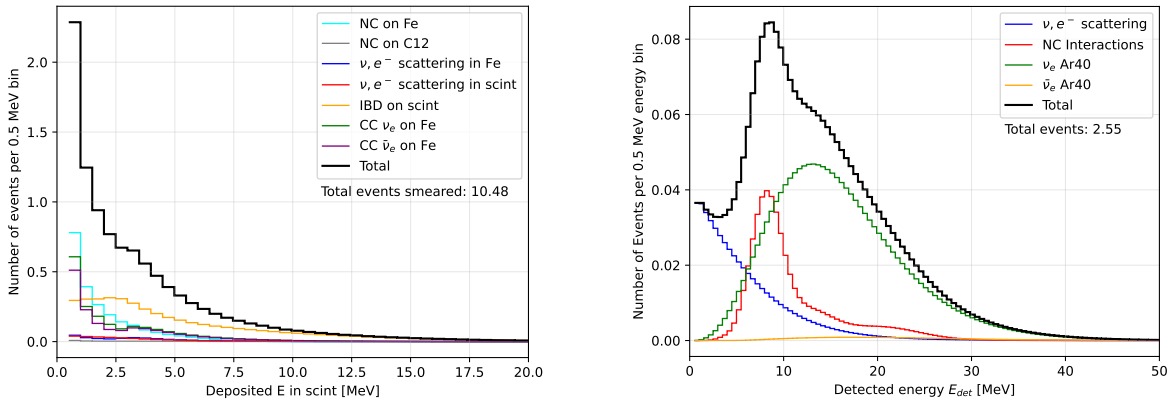
6 DUNE ND smeared SN event rates

Considering the detector smearing matrix R_{ij} and the post-smearing efficiency for the i-th reconstructed-energy bin, the smeared event rate for channel c and the i-th energy bin can be written as:

$$N^i_{\text{smeared, c}} = \epsilon_i \sum_j R_{ij} N^j_{\text{unsmeared, c}} \tag{7}$$

6.1 TMS supernova neutrino signal

Using the smearing matrices calculated in Section 5, a prediction for the neutrino signal for a SN at a distance of 10 kpc can be made. No post-smearing efficiencies have been added. Since the lowest SNOwGLoBES energy bin starts at 0.5 MeV, all events that deposit less than this threshold in the TMS scintillator are not included in the smeared signal.



(a) Smeared SN signal of TMS.

(b) Smeared SN signal of NDLAr.

Figure 5: Smeared signal comparison. Garching flux and SN distance of 10 kpc.

For the Garching flux, around 10.5 supernova neutrino events are expected to be registered by the TMS before any background contribution is included. On the other hand, using the Livermore flux, around 52.3 events are estimated.

6.2 ND-LAr smeared signal

The NDLAr detector was not simulated in Geant4. For the smearing matrices, the ar40kt response matrices were chosen as a simplified approximation. These files are provided in SNOwGLoBES. No additional post-smearing efficiencies were applied.

7 Cosmic muon background at TMS

7.1 Background model

The DUNE near detector complex is 60 m underground. For ND-LAr, a cosmic-ray rate of ≈ 100 Hz is expected [2]. Assuming that the cosmic-muon rate scales approximately with the detector top surface area, a rate of 150 Hz is estimated for the TMS since the top surface area of the TMS is 1.5 times that of ND-LAr. For a 9 s supernova neutrino signal, 1350 background events can be assumed. For simplicity, only cosmic muons are considered in this study. EcoMug is used to generate cosmic muons [23]. It simulates cosmic muons at the surface level. As a simplifying assumption, an energy loss of 36.2 GeV through 60 m of rock is assumed. With an approximate rock density of 2.7 $\frac{\mathrm{g}}{\mathrm{cm}^3}$, the column depth is $1.6 \times 10^4 \frac{\mathrm{g}}{\mathrm{cm}^2}$. Looking at the CSDA ranges from Ref. [24], an energy loss of 36.2 GeV is estimated. The cosmic muons are generated on a plane above the TMS Geant4 model. EcoMug provides the muon energy, momentum, and direction. The energy of the muon is approximated as follows:

$$E_\mu = E_0 - \frac{36.2\,\mathrm{GeV}}{\cos(\theta)} \tag{8}$$

θ represents the zenith angle of the cosmic muon. A larger angle leads to a longer path through the rock and, in a continuous approximation, to a higher energy loss.

7.2 Background rejection

As seen in Fig. 6, many cosmic muons deposit much more energy in the TMS scintillator compared to supernova neutrino events (Fig. 5a). Applying a simple energy cut where only energy deposits in the range $0.5\,\mathrm{MeV} \leq E_{\mathrm{scint}} \leq 15\,\mathrm{MeV}$ are considered, 54.8 background events remain in this energy interval. Additionally, cosmic muons tend to have longer path lengths in the scintillator. Applying an additional $L_{\mathrm{cut}} = 30\,\mathrm{mm}$, for example, would cut 67.6% of remaining background events and keep 81.3% of SN neutrino interactions. Applying both cuts, 8.3 SN events and 17.8 background events are estimated. Additional reconstruction information available in the full TMS detector may provide further background rejection beyond the observables considered here.

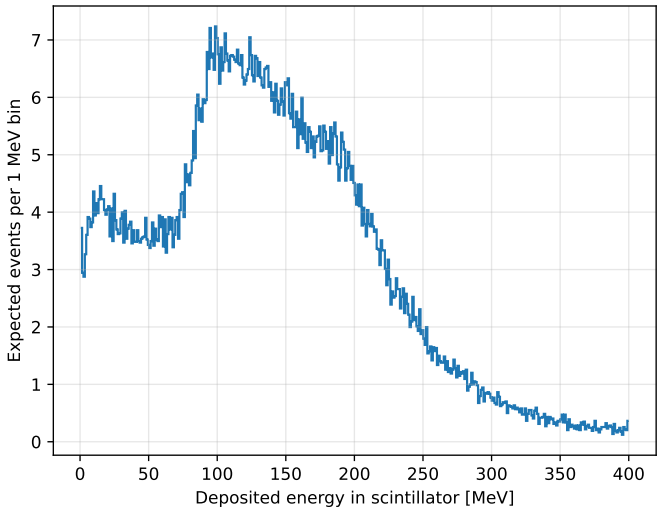


Figure 6: Cosmic-muon scintillator energy-deposition distribution.

Selection	Signal	Background	Signal acceptance	Background rejection
No selection	10.5	1350	100%	0%
$0.5\,\text{MeV} \leq E_{\text{scint}} \leq 15\,\text{MeV}$	10.27	54.8	98.0%	95.9%
E_{scint} cut + $L_{\text{scint}} \leq 30\,\text{mm}$	8.3	17.8	79.2%	98.7%

Table 3: SN events kept and background events rejected for different cuts.

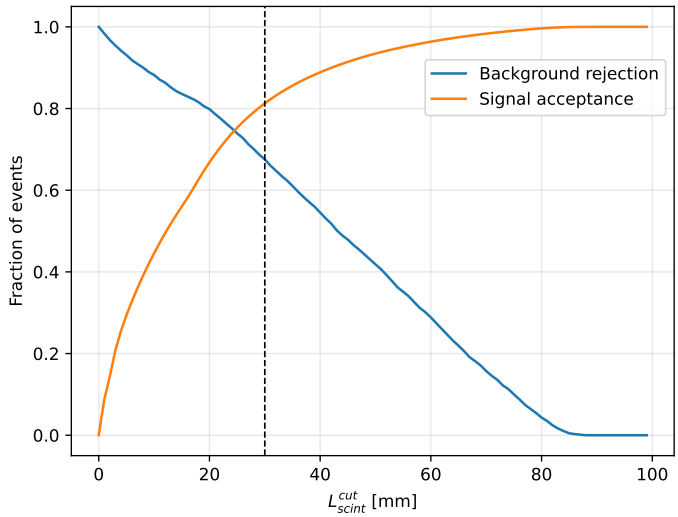


Figure 7: Fraction of background events cut and SN events kept. The dashed line shows an example cut of 30 mm.

8 Conclusion

The goal of this study was to estimate the sensitivity of the DUNE near detector complex to supernova neutrinos. Using SNOwGLoBES, unsmeared event rates were calculated for all near detector components for both the Garching and Livermore flux models. For the TMS specifically, a dedicated detector-response model was developed. MARLEY was used as an event generator, while Geant4 was used to model the TMS detector geometry and determine detector responses for several neutrino interaction channels. Using these detector-response matrices, 10.5 supernova-neutrino events are expected in the TMS for the Garching flux model and 52.3 events for the Livermore flux model for a supernova at a distance of 10 kpc. When cosmic-muon background is included and simple selection cuts are applied, 8.3 SN events for the Garching model and 17.8 cosmic-muon background events remain. These results indicate that the TMS may provide sensitivity to a Galactic supernova-neutrino burst, although the expected event rates depend strongly on the supernova distance and the assumed flux model. While cosmic-muon background remains relevant, simple observables such as the deposited scintillator energy and the path length in the scintillator can substantially reduce the background. A more realistic TMS reconstruction model could potentially provide further background rejection. For ND-LAr, the SNOwGLoBES ar40kt detector-response matrices were used as a first approximation. This results in approximately 2.6 expected SN events for the Garching flux model and 12.7 events for the Livermore flux model. This study therefore provides a first-order estimate of the supernova-neutrino sensitivity of the DUNE near detector complex. Several simplifying assumptions were made, including modeling the TMS steel absorber as pure ^{56}Fe and using a simplified muon-propagation model for the cosmic-background estimate. Radiological background, as well as detector noise backgrounds, are also neglected. A more complete sensitivity study would require realistic trigger and reconstruction models, a more detailed treatment of backgrounds, and a detector-response model for the remaining near detector components.

Acknowledgments

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